

ML PROJECT REPORT: COPD RISK

Team PMs – members:

- 1. Vaishak Prasad Bhat – IMT2023085
- 2. P Venkata Rami Reddy – IMT2023102
- 3. Harshit Krishna R – IMT2023613

Github link: <https://github.com/vaishak-iiitb/COPD.git>

Task:

The objective of this challenge is to build a machine learning model capable of predicting the risk of Chronic Obstructive Pulmonary Disease (COPD) in patients. This is a binary classification problem where the target variable is `has_copd_risk` (1 = COPD risk present, 0 = No COPD risk).

The aim is to maximize the **F1 Score**, a metric chosen specifically to balance precision and recall due to the potential class imbalance inherent in medical diagnosis datasets. The project involved training three distinct model architectures—Logistic Regression (with extensive feature engineering), Support Vector Machine (SVM), and Multi-Layer Perceptrons (Neural Networks)—to determine which method best captures the clinical and physiological indicators of COPD.

Dataset and Features:

The dataset contains clinical, physiological, and lifestyle measurements collected from patients. It consists of a training set of **44,553** records and a test set of **11,139** records. The data is relatively clean but requires significant preprocessing to handle skewed physiological distributions (like enzyme levels) and categorical encoding

Column1	Column2	Column3
Column Name	Type	Description
patient_id	Identifier	Unique identifier assigned to each patient record.
sex	Feature	Biological sex of the patient (M/F).
age_group	Feature	Patient age categorized in 5-year bands.
height_cm	Feature	Height of the patient in centimeters.
weight_kg	Feature	Weight of the patient in kilograms.
waist_circumference_cm	Feature	Waist circumference (cm); indicator of metabolic risk.
vision_left / right	Feature	Visual acuity measurements for left and right eyes.
hearing_left / right	Feature	Hearing test results for left and right ears.
bp_systolic / diastolic	Feature	Blood pressure values (mmHg).
fasting_glucose	Feature	Blood glucose concentration after fasting (mg/dL).
Lipid Profile	Features	Includes total_cholesterol, triglycerides, hdl_cholesterol, ldl_cholesterol.

hemoglobin_level	Feature	Hemoglobin concentration in blood (g/dL).
urine_protein_level	Feature	Urine protein presence level (kidney function indicator).
serum_creatinine	Feature	Concentration of creatinine in blood (mg/dL).
Enzymes	Features	Liver function markers: ast_enzyme_level, alt_enzyme_level, ggt_enzyme_level.
Oral Health	Features	oral_health_status, dental_cavity_status, tartar_presence.
has_copd_risk	Target	1 = COPD risk present, 0 = No risk.

Exploratory Data Analysis (EDA):

The EDA phase was conducted to understand the underlying structure of the medical data, detect anomalies, and identify the strongest predictors of COPD risk. The analysis revealed that while the dataset was syntactically clean, it contained significant physiological variance and class imbalance.

1. Dataset Structure and Target Balance

- **Dimensions:** The training dataset consists of 44,553 patient records with 25 potential feature columns. The test set contains 11,139 records.
- **Data Completeness:** An initial null-check confirmed that there were no missing values across any columns in the raw dataset (0.0% missingness). However, distinct placeholder values (e.g., vision scores of 9.9) were identified later as anomalies requiring treatment.
- **Class Imbalance:** The target variable `has_copd_risk` exhibits a moderate imbalance :
 - No Risk (0): 63.3%
 - Risk Present (1): 36.7%
 - This distribution (approx. 1.7:1 ratio) confirmed the necessity for class-weighting strategies in the models and the use of the F1-score rather than accuracy as the primary evaluation metric.

2. Univariate Analysis: Feature Distributions

Histograms and boxplots revealed three distinct categories of feature behaviors:

- **Normal-like Distributions:** Anthropometric features such as `height_cm` and `waist_circumference_cm` followed a roughly normal distribution, centered around 165cm and 82cm respectively. `age_group` showed multimodal peaks at 40 and 60, reflecting the 5-year band categorization.
- **Right-Skewed Distributions:** Physiological markers, particularly liver enzymes and lipids, showed heavy right tails (positive skew).
 - triglycerides, `alt_enzyme_level`, `ast_enzyme_level`, and `ggt_enzyme_level` all exhibited extreme outliers on the high end .
 - *Action:* This observation led to the application of Log1p transformations in the Neural Network pipeline to compress the dynamic range and stabilize gradient descent.
- **Discrete/Anomalous Distributions:**

- vision_left and vision_right were dominated by a spike at 1.0 (normal vision) but contained a distinct cluster of values at 9.9, which is clinically implausible and likely a code for "blindness" or "measurement failed".
- hearing_left and hearing_right were effectively binary (1 or 2), with the vast majority at 1 (normal hearing).

3. Bivariate Analysis: Correlation with Target A correlation analysis (using point-biserial correlation for the binary target) identified several key drivers of COPD risk.

- Top Positive Correlations:
 - Hemoglobin Level (): This was the strongest continuous predictor. High hemoglobin often correlates with COPD as the body compensates for low oxygen levels (polycythemia).
 - Height (): A surprising positive correlation, potentially confounding with sex (males tend to be taller and may have higher occupational exposure).
 - Weight () & Triglycerides (): Confirmed the link between metabolic syndrome factors and respiratory health .
- Top Negative Correlations:
 - Sex (): When encoded (M=0, F=1), the negative correlation indicates that Males in this dataset have a significantly higher risk incidence than Females.
 - HDL Cholesterol (): Higher "good" cholesterol was associated with lower COPD risk.

4. Feature Engineering Insights The EDA suggested that isolated features were less predictive than their interactions.

- Metabolic Syndrome: Variables like waist_circumference_cm and fasting_glucose had weaker individual correlations () compared to their combined impact. This justified the creation of a Metabolic Score and lipid ratios (e.g., total_cholesterol / hdl_cholesterol) during the feature engineering phase.
- Outlier Management: Boxplots highlighted that enzyme levels could reach into the thousands (e.g., ALT max ~2914) , necessitating Winsorization (capping at the 99th percentile) to prevent these extreme values from distorting linear model coefficients.

Data Preprocessing:

1. Advanced Feature Engineering For models incapable of learning non-linear interactions automatically (Logistic Regression), we manually engineered features. This included:

- BMI and Waist-to-Height Ratio: Calculated to capture obesity risks.
- Interaction Terms: Pairwise multiplications of highly correlated features (e.g., hemoglobin_level * height_cm).
- Metabolic Scores: A composite score derived from triglycerides, glucose, and BMI.

2. Transformation and Scaling

- Log Transformation: Applied np.log1p to skewed enzyme and lipid columns to compress their dynamic range.

- **Scaling:** StandardScaler was used for the majority of features. A dual-scaling strategy (RobustScaler for outliers, StandardScaler for others) was employed for the Neural Network to improve gradient stability.

3. **Categorical Encoding** Binary variables (sex, oral_health_status) were mapped to 0/1. Multi-class categoricals were one-hot encoded for Logistic Regression and label-encoded for tree-based models or deep learning embeddings.

Models used for training:

To tackle the binary classification task of predicting COPD risk, the project systematically evaluated three distinct modeling approaches, ranging from linear baselines to complex deep learning architectures. A Logistic Regression model served as the initial baseline, utilizing ElasticNet regularization to perform feature selection and prevent overfitting on a manually expanded feature set of 115 interaction terms. To better capture non-linear physiological relationships without explicit feature engineering, a Support Vector Machine (SVM) was employed with a Radial Basis Function (RBF) kernel and class weighting to penalize misclassifications of the minority risk class. The most advanced model was a deep Multi-Layer Perceptron (Neural Network) featuring a tapering dense architecture (512 to 256 to 128 units) with Batch Normalization and Dropout layers. This neural network proved superior by automatically learning latent representations of metabolic syndrome factors from robustly scaled and Log1p-transformed inputs, ultimately achieving the highest F1 score through a custom threshold optimization loop.

1. Logistic Regression (ElasticNet)

We utilized Logistic Regression as a baseline, but enhanced it significantly to handle the complexity of medical data.

Methodology: To capture non-linearities, we expanded the feature space to 115 features using interaction terms. We used ElasticNet regularization (penalty='elasticnet') to perform automatic feature selection and prevent overfitting on this expanded set.

Calibration: The raw probabilities were calibrated using Isotonic Regression to ensure the output probabilities reflected true risk rates.

Performance:

Validation F1: 0.672 (at 0.5 threshold).

Optimized F1: 0.7134 (at threshold 0.300).

2. Support Vector Machine (SVM)

The SVM was deployed to capture non-linear decision boundaries using the Radial Basis Function (RBF) kernel, which projects data into higher-dimensional space without manual feature expansion.

Methodology: We utilized class_weight='balanced' to penalize misclassifications of the minority class (COPD cases) more heavily. The regularization parameter C was tuned to 2.0.

Performance:

Validation ROC AUC: 0.8426.

Optimized F1: 0.7193 (at threshold 0.340).

The SVM outperformed the Logistic Regression, proving that the non-linear kernel was more effective than manual interaction engineering.

3. Neural Network (Multi-Layer Perceptron)

We designed a Deep Learning model to automatically learn hierarchical representations of the medical data. This model was the primary focus of our experimentation.

Architecture: A deep architecture of decreasing density (512 to 256 to 128 nodes) with ReLU activation.

Regularization: We implemented Batch Normalization and Dropout (0.2 - 0.3) after each dense layer to prevent overfitting.

Optimization: The model used the Adam optimizer with a ReduceLROnPlateau scheduler. Crucially, we implemented a custom threshold tuning loop on Out-Of-Fold (OOF) predictions to maximize F1.

Performance:

Mean CV F1: 0.7203.

The Neural Network achieved the highest performance metrics across all experiments.

Discussion on the Performance of Different Approaches:

To effectively model the binary risk of COPD, the project implemented and compared three distinct classes of algorithms: Linear Models (Logistic Regression), Kernel-based Models (Support Vector Machine), and Deep Learning (Multi-Layer Perceptrons). The performance hierarchy clearly demonstrated the value of model complexity when dealing with physiological data.

1. Logistic Regression (Baseline & Enhanced):

The initial approach utilized Logistic Regression, a linear classifier valued for its interpretability. However, given the complex, non-linear interactions typical in medical data (such as the compounded risk of age and BMI), a standard linear model was insufficient. To mitigate this, we employed an extensive Elastic Net strategy, which combines L1 and L2 regularization to perform feature selection while preventing overfitting. We also manually expanded the feature space to 115 variables by engineering interaction terms. Despite this heavy manual engineering, the model achieved a Validation F1 score of 0.672 at the default threshold, which improved to 0.7134 only after aggressively lowering the decision threshold to 0.300. This indicated that the linear decision boundary struggled to separate the classes without sacrificing significant precision.

2. Support Vector Machine (SVM):

The SVM model represented a step up in complexity, utilizing a Radial Basis Function (RBF) kernel to project the features into a higher-dimensional space where they might be linearly separable. Unlike the Logistic Regression model, the SVM did not require the manual creation of interaction features to find non-linear patterns. By setting the regularization parameter to 2.0 and using balanced class weights to penalize minority class errors, the SVM effectively created "risk zones" in the data. This approach yielded a Validation F1 of 0.7193 at an optimized threshold of 0.340. The jump in performance over Logistic Regression confirmed that the underlying risk factors interact in non-linear ways that a kernel method can capture more naturally than a manually engineered linear model.

3. Multi-Layer Perceptron (Neural Network):

The Deep Learning approach utilized a Neural Network with a tapering architecture (512 to 256 to 128 neurons) designed to compress the input features into latent representations of patient health.

This model proved to be the most robust. By incorporating specific preprocessing steps like Log1p transformations for skewed enzymes and Robust Scaling, the network was able to handle outliers better than the SVM or Logistic Regression. The network achieved a Mean Cross-Validation F1 of 0.7203. While the performance gain over SVM was marginal, the Neural Network demonstrated superior stability across folds and required less manual feature engineering to achieve its results.

Our interpretation of why Ensemble Classifier Gave the Highest Accuracy:

The superior performance of the Multi-Layer Perceptron (F1: 0.720), and the general "ceiling" observed across all models around the 0.72 mark, leads to several critical interpretations regarding the dataset and the nature of COPD prediction.

1. The "Metabolic Syndrome" Latent Factor The success of the Neural Network is largely attributed to its ability to internally learn a representation of "metabolic syndrome." In the Logistic Regression model, we had to explicitly tell the model to look at specific ratios, such as waist_circumference / height or triglycerides * hemoglobin, to capture risk factors. The Neural Network, through its hidden layers, automatically inferred these complex non-linear relationships without manual instruction. The strong correlations found in the EDA between COPD risk and factors like weight, triglycerides, and hemoglobin suggest that the model is effectively acting as a metabolic disorder detector. Since metabolic stress is a known comorbidity and risk factor for COPD, the neural network was able to exploit this latent signal more effectively than the linear models.

2. Threshold Sensitivity and Class Imbalance A defining characteristic of all three successful models was their reliance on shifted decision thresholds. None of the models performed optimally at the default 0.5 probability cutoff.

Logistic Regression required a threshold of 0.300.

SVM required 0.340.

Neural Network required ~0.475.

This consistent downward shift indicates that the "signal" for COPD in this dataset is subtle. The models are generally conservative, often outputting low probabilities (e.g., 35% or 40%) even for positive cases. To effectively capture the minority class (Recall) without destroying Precision, we had to accept these lower confidence predictions as "Risk Present." The fact that the Neural Network's optimal threshold was closest to 0.5 suggests it was the best calibrated model, distinguishing "Risk" from "No Risk" with greater statistical confidence than the SVM or Logistic Regression.

3. The "Bayes Error" Limit The fact that a highly complex Neural Network (0.720) only marginally outperformed a well-tuned SVM (0.719) and an engineered Logistic Regression (0.713) suggests we hit a hard data limitation. We likely reached the "Bayes Error" rate—the lowest possible error rate achievable for this specific set of input features. The dataset contains standard vitals (blood pressure, height, weight) and blood panels, but it lacks specific respiratory metrics (such as spirometry data) or behavioral history (such as smoking status), which are the clinical gold standards for COPD diagnosis. Without these domain-specific features, the models are effectively predicting "general poor health" or "metabolic stress" as a proxy for COPD. This proxy

relationship limits the maximum achievable F1 score to the low 0.70s, regardless of how complex the algorithm becomes.

Interesting Observations:

The machine learning process yielded several key observations, primarily focused on the predictive limitations of the features and the necessary interventions required for model stability.

1. **The Performance Plateau and Data Limits:** A significant finding was the consistent clustering of F1 scores around 0.72 across all models, ranging from the engineered Logistic Regression (0.713) to the complex Neural Network (0.720). This suggests that the predictive capacity is limited by the feature set itself, reaching the Bayes Error rate achievable with standard clinical and blood panel data. The models are likely predicting "general poor health" as a proxy for COPD, and no amount of algorithmic complexity can extract signal for COPD that is missing (such as actual spirometry results or detailed smoking history).
2. **Extreme Threshold Sensitivity:** A defining characteristic across all successful models was the need for a manually optimized decision threshold, which was consistently lower than 0.5 (ranging from 0.300 to 0.475). The models were fundamentally conservative, rarely outputting high probabilities even for positive cases. This sensitivity highlights the dataset's mild class imbalance and the necessity to deliberately prioritize Recall by accepting lower-confidence predictions to effectively capture the minority high-risk class.
3. **Strong Demographic Signal:** The biological sex of the patient emerged as the single most strongly correlated demographic factor (correlation ≈ -0.51). This finding indicates a significant disparity in risk prevalence, where males in this dataset exhibited a much higher risk of COPD than females. The models heavily relied on this powerful demographic signal, a critical observation for interpretability and fairness analysis.
4. **Physiological Feature Redundancy:** Analysis of the extensive feature set demonstrated rampant multicollinearity. The aggressive feature pruning necessary during regularization often eliminated close to 100 highly correlated variables, leaving only the most distinct physiological signals. This highlights the high degree of redundancy inherent in standard medical lab panels, where many different measurements (like LDL and Total Cholesterol) convey essentially the same metabolic information.
5. **Non-Standard Normalization Requirements:** Unlike basic machine learning tasks, this physiological dataset required specialized data preparation beyond standard scaling. Specifically, highly skewed lab values like liver enzymes (ALT, GGT) and lipids required the Log1p transformation to compress their dynamic range and reduce the distorting influence of extreme outliers. Without this specific normalization, the gradients of the Neural Networks were unstable, and the performance of all models was compromised.

Impact of Training Data Subsampling (F1 score analysis):

Reducing the training dataset size by 80% resulted in a decrease in predictive performance for both the Multi-Layer Perceptron (MLP) and the Support Vector Machine (SVM) models.

Model	F1 Score (Full Data)	F1 Score (20% subsample)	F1 Score (10% subsample)
MLP	0.7190	0.7081	0.7031
SVM (RBF Kernel)	0.7193	0.7090	0.7158

1. The Narrow Competitive Margin (100% Data)

At full data capacity, the minute F1 margin between the SVM (0.7193) and the MLP (0.7190) indicates Feature Learning Parity.

- **Data Saturation:** Both complex models successfully extracted the maximum predictive signal available from the core clinical and metabolic features in the dataset. This suggests they both hit a performance ceiling imposed by the Bayes Error rate—the best possible result given the limited fidelity of the input data (which lacks gold-standard diagnostic information).
- **Optimal Learning Alignment:** The MLP's ability to learn optimal, non-linear feature representations resulted in a final classification function almost identical to the function found by the SVM's generic but powerful RBF kernel.

2. Performance Degradation on Subsample (20% Data)

The drop in performance for both models is approximately 1%, but this drop arises from fundamentally different constraints:

A. Multi-Layer Perceptron (MLP): Parameter Starvation

The MLP is a classic data-hungry model. Its substantial number of parameters (weights) distributed across its deep layers requires a massive volume of unique training examples to stabilize.

- When restricted to the 20% subsample, the model lacked sufficient data to fully explore and settle into the optimal weight configuration. This led to increased variance and resulted in the noticeable 1.09% performance drop.

B. Support Vector Machine (SVM): Boundary Integrity Failure

The SVM is theoretically not data-hungry because it focuses only on the boundary points. However, its decline is due to a failure to maintain the reliability of the classification boundary under reduced input.

- **Boundary Robustness Degradation:** In complex, high-dimensional spaces—like the one created by the RBF kernel using noisy health data—the 20% subsample did not provide a reliable set of support vectors.
- **Kernel Instability:** The resulting maximal margin hyperplane found by the SVM became highly sensitive to the few noisy points remaining in the small sample. This instability meant the learned boundary was less generalizable when applied to the full variance of the test set, leading to its performance decline.